INDIVIDUAL WEEKLY PROJECT REPORTS

Student: Annalakshmi R

Role: Data Preprocessing and Exploratory Data Analysis (EDA)
Project Title: Flight Delay Analysis and Visualization

WEEK 3 – Data Cleaning and Preprocessing

During the third week, I focused on the data cleaning and preprocessing stage. This was an important part of the project because the quality of the dataset directly influences the reliability of the analysis. I worked on examining the dataset for missing values, duplicate records, invalid entries, and inconsistencies.

I understood that missing values could affect calculations and visualizations if they were not handled appropriately. Therefore, I examined the available information to identify columns that required further attention. I also considered the importance of checking duplicate records because repeated entries could influence the results and create misleading patterns.

One of the important observations during this stage was the presence of negative values in the CRS elapsed time field. Since elapsed time represents the duration between two events, negative values required investigation. I considered these entries as potential data quality issues and discussed their treatment with the team.

I also focused on understanding the importance of maintaining appropriate data types. Numerical variables needed to be represented correctly for calculations, while categorical variables such as airline identifiers required suitable formats for grouping and comparison.

I coordinated with the data collection team to ensure that the identified issues were properly understood before proceeding with further analysis.

Outcome and Learning:

This week helped me understand practical data cleaning techniques and the importance of validating raw data. I learned that preprocessing is essential for reducing errors and improving the quality of analytical results.